

Mukesh Arambakam

(+353) 89-4960-450 | amukesh.mk@gmail.com | Stamp 4 | <https://linkedin.com/in/mukeshmk95/> | <https://mukeshmk.github.io/>

Personal Summary

Senior Machine Learning Engineer with with **7+ years** of Software Development experience, including **5 years** specializing in **Natural Language Processing, Neural Machine Translation**, and **Generative AI**, backed by a Master's in Data Science. Skilled in designing and delivering end-to-end ML solutions, including Data Preprocessing, Model Training, Evaluation and Optimization. Experience in fine-tuning LLMs with PEFT methods, building **Agentic AI** systems, and developing scalable, production-ready ML applications. Strong background in **Machine Learning, Artificial Intelligence, Statistical Modeling**, and **Data Analytics**.

Skills

Programming Languages: Python, Java, R, JavaScript, C++, SQL, PL/SQL

Machine Learning Libraries: HuggingFace, PyTorch, NLTK, spaCy, transformers, OpenNMT, scikit-learn, TensorFlow, fasttext

Generative AI & LLM Technologies: LangChain, LangGraph, FastMCP, Agentic AI, Tool Calling, Prompt Engineering, RAG, LLM-as-a-Judge

LLM Fine-Tuning & Optimization: PEFT, LoRA (Low-Rank Adaptation), QLoRA (Quantized LoRA), In-Context Learning, ONNX

Other Python Libraries: NumPy, SciPy, Pandas, FastAPI, matplotlib, elasticsearch, fastalign

Infrastructure & DevOps: Docker, Docker Compose, Jenkins, git, JIRA, Oracle DB, Shell

Work Experience

Oracle Corporation

SENIOR MACHINE LEARNING ENGINEER (IC-3)

Nov. 2020 - Current

LLMs & Agentic AI Systems:

- Launched an agentic AI system with LangGraph, supporting 10k+ LLM calls to Translation & LangDetect MCP servers.
- Implemented 2 RAG systems with vector databases for domain-specific data extraction and translation enhancement.
- Engineered 5+ prompt engineering solutions for back-translation, domain classification, post-editing, and data generation with sophisticated tool calling mechanisms.
- Used prompt chaining techniques for multi-step LLM workflows including translation evaluation, iterative improvement, pivot-translation and post-editing.

Data Processing & Analysis

- Developed a dataset analysis framework from scratch for MT training data statistics, working across 40+ language pairs.
- Built a monolingual data generator for low-resource languages increasing synthetic data coverage by 30–40%.
- Designed a cross-lingual vocabulary gap detection tool leveraging n-gram and embeddings, improving coverage by 20%.
- Enhanced data-cleaning by reducing noise by 10% using alignment models, length ratios, and toxicity checks.
- Used in-context learning methods: zero-shot and few-shot learning for domain classification & data categorization tasks.

Engine Training, Fine-Tuning & Optimization:

- Updated the Training pipeline to generate word alignment, length ratio models for data cleaning and evaluations.
- Developed a Monolingual Engine Training pipeline and de-prioritized the tagged monolingual data during training.
- Fine-tuned translation models using PEFT techniques for efficient adaptation to domain-specific tasks.
- Applied quantization strategies (QLoRA) for LLM deployment, cutting resource usage by 40% in production.

Advanced Evaluation & Quality Assurance

- Re-engineered translation evaluation repository with modern architectures, boosting performance by 25%.
- Designed custom translation quality rubrics with linguists and applied LLM-as-a-Judge via direct assessment.
- Integrated G-Eval metric and LLM-as-a-Judge framework for comprehensive LLM-based translation assessment.
- Created visualization for evaluation scores using custom-built tools.
- Expanded evaluation framework with 5 new metrics: Comet, XComet, BERTScore, CometQE, and length-ratio checks.
- Assessed LLM translation consistency with Translate-then-Evaluate and LLM-as-a-Judge, raising accuracy by 20%.

Language Detection Enhancements:

- Expanded language detection coverage from 45 to 192 languages.
- Introduced a 3-stage hierarchical language detection architecture to differentiate sub-cultural languages.
- Improved language detection accuracy by 90% for very short strings (<12 characters).
- Refactored the codebase using robust system design patterns and FastAPI to significantly boost performance by 3x times.
- Packaged and published ONNX models with architecture-aware configurations for cross-compatibility.
- Used architecture-specific optimizations and model quantization to reduce memory footprint by 50%.

APPLICATIONS ENGINEER (IC-2)

June 2019 - Aug. 2019

- Coordinated with multiple global teams, business analysts, and stakeholders to develop a cross-platform application.
- Lead developer in overhauling the project security flow of the Product and introduced 2 new features.
- Implemented a micro-service which communicates between tenants in a *Multi-Tenant Architecture*.
- Mentored 10 junior team members, and provided insights about the product's internal working and underlying tools.

APPLICATIONS DEVELOPER (IC-1)

July 2017 - May 2019

- **Certificate of Appreciation:** For *outstanding contribution* to the Product's Development.
- Assumed a pivotal role in the development of 3 core features of the product along with my team members.
- Optimized the performance of an internal module's filter framework with the use of dynamic SQL queries by 30%.
- Improved the performance of *search framework* by 20%, using query optimization techniques.

Education

Trinity College of Dublin

1:1

M.SC. IN DATA SCIENCE

2019 - 2020

Thesis: **Federated Meta-Learning:** A Novel Approach for Algorithm Selection.

FMLearn, an application that allows the exchange of meta-data via APIs about machine learning models and data for the purpose of meta-learned algorithm selection and configuration.

Amrita School of Engineering

CGPA: 8.37/10.0

B.TECH. IN COMPUTER SCIENCE AND ENGINEERING

2013 - 2017

Publication

Jun 2020 **7th ICML Workshop on Automated Machine Learning (AutoML)**,

Federated Meta-Learning: Democratizing Algorithm Selection Across Disciplines and Software Libraries

Personal Projects

Home Lab Infrastructure

- Built complex home lab environment using Proxmox virtualization platform with Docker LXC's.
- Self-host comprehensive ecosystem supporting 10+ applications (Home Assistant, Nextcloud, Immich, Jellyfin, etc.)
- Implemented enterprise-grade authentication using Authelia SSO solution.
- Manage network infrastructure with custom DNS and DHCP servers.
- Secured remote access through Tailscale VPN and Funnels for all critical services.
- Developed complex home automation using Home Assistant for smart home integration.

Image and Audio Captcha Solver

- Parallelised captcha generator and solver with 93% accuracy built using TensorFlow's CNN model.

Connect 4 - based on Artificial Intelligence

- A multi-agent implementation of the game Connect-4 using MCTS, Minimax and Expectimax algorithms.